

Problem

When the total charge in a capacitor is doubled, the energy stored:

- (a) remains the same
- (b) is halved
- (c) is doubled
- (d) is quadrupled

Step-by-step solution

Step 1 of 1

The expression for the energy stored in the capacitor is,

$$w = \frac{q^2}{2C}$$

Consider that the total charge in the capacitor is doubled.

Find the new value of energy stored in the capacitor.

$$\begin{aligned} w_n &= \frac{(2q)^2}{2C} \\ &= \frac{4q^2}{2C} \\ &= 4w \end{aligned}$$

Hence, it is observed that the new value of energy is 4 times (quadrupled) greater than the normal value.

Therefore, the correct option is (d).